



Journal of Systems and Software

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About the journal

Aims and scope

For JSS's full CFP including information on Special Issues, Industry, Trends, and Journal First tracks please continue to read for further details.

The *Journal of Systems and Software* publishes papers covering all aspects of software engineering. All articles should provide evidence to support their claims, e.g. through empirical studies, simulation, formal proofs or other types of validation. Submissions should contribute new knowledge for software engineering research and practice, independently of specific application domains. Topics of interest include, but are not limited to:

- Methods and tools for software requirements, design, architecture, verification and validation, testing, maintenance and evolution
- Agile, model-driven, service-oriented, open source and global software development
- Human/social aspects in software engineering and developer experience
- Artificial Intelligence, data analytics and big data applied in software engineering
- Metrics and evaluation of software development resources
- DevOps, continuous integration, build and test automation
- Software Engineering education
- Ethical/societal aspects of Software Engineering
- Software Engineering for AI systems

- Software Engineering for Sustainability
- Methods and tools for empirical software engineering research

The journal welcomes reports of practical experience for all of these topics, as well as replication studies and studies with negative results. The journal appreciates the submission of systematic literature reviews, mapping studies and meta-analyses. However, these should report interesting and important results, rather than merely providing statistics on publication year, venue etc.

Open Science:

JSS supports Open Science and reproducible research through the JSS Open Science Initiative. For papers where authors indicate participation, the manuscript is automatically forwarded to the JSS Open Science Board, chaired by [Christoph Treude](#), after acceptance. Following successful review of availability and usability (which does not affect acceptance of the paper), the publisher will add a statement acknowledging validation of the Open Science material and include the paper in JSS's [Open Science collection](#). If material cannot be shared (e.g., due to NDAs), authors should state this explicitly. Depending on the study, Open Science material may include study protocols, (anonymized) data, analysis scripts, source code, tools, or codebooks.

In addition to regular papers, JSS features two special tracks (In Practice, New Ideas and Trends Papers), as well as special issues.

In Practice is exclusively focused on work that increases knowledge transfer from industry to research. It accepts: (1) Applied Research Reports where we invite submissions that report results (positive or negative) concerning the experience of applying/evaluating systems and software technologies (methods, techniques and tools) in real industrial settings. These comprise empirical studies conducted in industry (e.g., action research, case studies) or experience reports that may help understanding situations in which technologies really work and their impact. Submissions should include information on the industrial setting, provide motivation, explain the events leading to the outcomes, including the challenges faced, summarize the outcomes, and conclude with lessons learned, take-away messages, and practical advice based on the described experience. Contributing authors from industry are encouraged but not mandatory. (2) Practitioner Insights where we invite experience reports showing what actually happens in practical settings, illustrating the challenges (and pain) that practitioners face, and presenting lessons learned. Problem descriptions with significant details on the context, underlying causes and symptoms, and technical and organizational impact are also welcome. Practitioner insights papers may also comprise invited opinionated views on the evolution of chosen topic areas in practice. In contrast to applied research reports, practitioner insights are limited to four pages and the first author must be from industry. Finally, submissions to this track should be within scope of the journal's above topics of interest and they will be evaluated through industry-appropriate criteria for their merit in reporting useful industrial experience rather than in terms of academic novelty of research results.

New Ideas and Trends Papers

New ideas, especially those related to new research trends, emerge quickly. To accommodate timely dissemination thereof, JSS introduces the New Ideas and Trends Paper (NITP). NITPs should focus on the systems/software engineering aspects of new emerging areas, including: the internet of things, big data, cloud computing, software ecosystems, cyber-physical systems, green/sustainable systems, continuous software engineering, crowdsourcing, and the like. We distinguish two types of NITPs:

- A short paper that discusses a single contribution to a specific new trend or a new idea.
- A long paper that provides a survey of a specific trend, as well as a (possibly speculative) outline of a solution.

NITPs are not required to be fully validated, but preliminary results that endorse the merit of the proposed ideas are welcomed.

We anticipate revisiting specific new trends periodically, for instance through reflection or progress reports. New Ideas and Trend Papers warrant speedy publication.

Special Issue proposals

To submit a proposal for a special issue please submit your proposal [here](#) to Special Issues Editors [Prof. Raffaella Mirandola](#) and [Prof. Laurence Duchien](#). Please visit the [special issue guidelines](#) page first to review the proposal guidelines and to download the proposal template required when submitting a proposal.

Journal First Initiative

Authors of JSS accepted papers have the opportunity to present their work in those conferences that offer a Journal First track. Using this track, researchers may take the best from two worlds: ensuring high quality in the JSS publication (thorough, multi-phase review process of a long manuscript), while getting feedback from a community of experts and fostering possible collaborations during a scientific event.

Details may vary from conference to conference, but generally speaking, JSS papers to be presented in a Journal First track must report completely new research results or present novel contributions that significantly extend previous work. The ultimate decision to include a paper in the conference program is up to the conference chairs, not JSS. A JSS paper may be presented only once through a Journal First track.

As of today, the list of conferences with which JSS is collaborating, or has collaborated, through a Journal First track, is: ASE, ICSME, SANER, RE, ESEM, PROFES, and APSEC.

Peer review

This journal follows a single anonymized review process. Your submission will initially be assessed by our editors. If your submission is deemed suitable, it will typically be sent to be reviewed by a minimum of two reviewers for an independent expert assessment of the scientific quality. The decision as to whether your article is accepted or rejected for publication will be made by our editors.

Read more about [peer review](#).

Our editors are not involved in making decisions about papers which:

- they have written themselves.
- have been written by family members or colleagues.
- relate to products or services in which they have an interest.

Any such submissions will be subject to the journal's usual procedures and peer review will be handled independently of the editor involved and their research group. Read more about [editor duties](#).

Authors may submit a formal appeal request to the editorial decision, provided it meets all the requirements and follows the procedure outlined in Elsevier's [Appeal Policy](#). Only one appeal per submission will be considered and the appeal decision will be final.

Special issues and article collections

The peer review process for special issues and article collections follows the same process as outlined above for regular submissions, except, a guest editor may send the submissions out to the reviewers and may recommend a decision to the journal editor. The journal editor oversees the peer review process of all special issues and article collections to ensure the high standards of publishing ethics and responsiveness are respected and is responsible for the final decision regarding acceptance or rejection of articles.

Open access

We refer you to our [open access information page](#) to learn about open access options for this journal.

Ethics and policies

Ethics in publishing

Authors must follow ethical guidelines stated in Elsevier's [Publishing Ethics Policy](#).

Submission declaration

When authors submit an article to an Elsevier journal it is implied that:

- The work described has not been published previously except in the form of a preprint, an abstract, a published lecture, academic thesis or registered report. See our policy on [multiple, redundant or concurrent publication](#).
- The article is not under consideration for publication elsewhere.

- The article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
- If accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright-holder.

To verify compliance with our journal publishing policies, we may check your manuscript with our screening tools.

Authorship

All authors should have made substantial contributions to all of the following:

1. The conception and design of the study, or acquisition of data, or analysis and interpretation of data
2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

All authors are accountable for all aspects of the work, to ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Corresponding author

Authors must appoint a single corresponding author to communicate with the journal during the editorial process. This individual submits the manuscript and all required documentation to the journal, and serves as the primary contact between the journal and the co-authors. It is important that all contact details of the corresponding author are kept up to date throughout the submission and publication process.

Only the corresponding author's affiliation will be used to determine eligibility for a publishing agreement, and possible discounts related to it. Affiliations of other co-authors are not relevant for eligibility.

Group authorship

Group authorship refers to the attribution of an article to a defined and named group of authors whose members have contributed to the work. Examples include research consortia, clinical trial groups, collaborative networks, task forces or working groups, expert panels or consensus groups, and participatory research groups.

- Group authorship must be declared on the title page *at submission*. When available, the submitting author should include the group under "Collaborative Author Groups" in the submission system.
- A corresponding author who agrees to coordinate all communications on behalf of the group must be designated.
- The group name will be included in the author list in the published article (e.g., "on behalf of the [Consortium Name]").
- All group members associated with the work should be listed in a dedicated section, preferably an appendix.
- All listed authors must follow the journal's authorship and authorship change policies.
- The group must maintain a transparent internal process for determining authorship and resolving disputes. All authorship disputes must be resolved within the group.

Changes to authorship

The editors of this journal generally will not consider changes to authorship once a manuscript has been submitted. It is important that authors carefully consider the authorship list and order of authors and provide a definitive author list at submission.

The policy of this journal around authorship changes:

- All authors must be listed in the manuscript and their details entered into the submission system. Changes can only be made prior to acceptance, and only if approved by the journal editor. This includes additions, deletion, or rearrangement of author names.

- Requests to change authorship must be made by the corresponding author using the [Authorship Change Request form](#). The corresponding author must provide the reason for the request to the journal editor with written confirmation from all authors, including any authors being added or removed, that they agree with the changes. Requests which do not comply with the instructions outlined in the form will not be considered.
- This journal does not allow authorship changes after acceptance. This includes additions, deletions, or the rearrangement of author names, including changes to the corresponding author.
- The review process may be paused while a change in authorship request is being considered.
- Change requests made by the authors without the [Authorship Change Request form](#) or editor approval may result in the rejection of the manuscript, or retraction if the article has already been published.

Declaration of competing interests

All authors must disclose any financial and personal relationships with other people or organizations that could inappropriately influence or bias their work. Examples of potential competing interests include:

- Employment
- Consultancies
- Stock ownership
- Honoraria
- Paid expert testimony
- Patent applications or registrations
- Grants or any other funding
- Affiliation with the journal as an Editor or Advisory Board Member

The [declarations tool](#) should always be completed.

Authors with a journal affiliation to declare should enter the following text under “Other Activities” within the [declarations tool](#) and should inform the journal and publisher prior to completing the submission process:

Given their role as [AUTHOR JOURNAL ROLE TITLE], [AUTHOR NAME] had no involvement in the peer-review of this article and has no access to information regarding its peer-review. Full responsibility for the editorial process for this article was delegated to another journal editor.

Editorial disclosure statements will be included as a footnote and/or in the declaration of competing interest section of the article.

Authors with no competing interests to declare should select the option "I have nothing to declare".

The resulting Word document containing your declaration should be uploaded at the "attach/upload files" step in the submission process. It is important that the Word document is saved in the .doc/.docx file format. Author signatures are not required.

Funding sources

Authors must disclose any funding sources who provided financial support for the conduct of the research and/or preparation of the article. The role of sponsors, if any, should be declared in relation to the study design, collection, analysis and interpretation of data, writing of the report and decision to submit the article for publication. If funding sources had no such involvement this should be stated in your submission.

List funding sources in this standard way to facilitate compliance to funder's requirements:

Funding: This work was supported by the National Institutes of Health [grant numbers xxxx, yyyy]; the Bill & Melinda Gates Foundation, Seattle, WA [grant number zzzz]; and the United States Institutes of Peace [grant number aaaa].

It is not necessary to include detailed descriptions on the program or type of grants, scholarships and awards. When funding is from a block grant or other resources available to a university, college, or other research institution, submit the name of the institute or organization that provided the funding.

If no funding has been provided for the research, it is recommended to include the following sentence:

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI use

Authors must declare the use of generative AI tools in the manuscript preparation process upon submission of the paper.

Generative AI and AI-assisted technologies ("AI tools") have been rapidly adopted in research and scholarly publishing, offering significant benefits while requiring clear standards for responsible and transparent use.

Authors preparing a manuscript for this journal may use AI tools to support them. However, these tools must never be used as a substitute for human critical thinking, expertise and evaluation. AI tools may only be applied with human oversight and control.

Please read Elsevier's detailed author policy on the use of AI tools, as stated in Elsevier's [generative AI policies for journals](#).

Ultimately, authors are responsible and accountable for the contents of their work. This includes accountability for:

- Carefully reviewing and verifying the accuracy, comprehensiveness, and impartiality of all AI-generated output (including checking the sources, as AI-generated references can be incorrect or fabricated).
- Editing and adapting all material thoroughly to ensure the manuscript represents the author's authentic and original contribution and reflects their own analysis, interpretation, insights and ideas.
- Ensuring accuracy and originality of any AI generated images and provide appropriate attribution where relevant.
- Ensuring the use of any tools or sources, AI-based or otherwise, is made clear and transparent to readers. If AI tools have been used, we require a disclosure statement upon submission; please see example below.
- Ensuring the manuscript is developed in a way that safeguards data privacy, intellectual property and other rights, by checking the terms and conditions of any AI tool that is used.

Finally, authors must not list or cite AI tools as an author or co-author on the manuscript since authorship implies responsibilities and tasks that can only be attributed to, and performed by, humans.

The use of AI tools in the manuscript preparation process must be declared by adding a statement at the end of the manuscript when the paper is first submitted. The statement will appear in the published work and should be placed in a new section before the references list. An example:

Title of new section: *Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.*

Statement: *During the preparation of this work the author(s) used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.*

The declaration does not apply to the use of basic tools, such as tools used to check grammar, spelling and references. In addition, authors are not required to disclose the use of AI tools or features within specialist disability-related assistive technology, provided these are used solely for accessibility purposes.

If you have nothing to disclose, you do not need to add a statement.

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In support of [Open Science](#), this journal offers its authors a free preprint posting service on [SSRN](#), Elsevier's preprint and early research repository.

During submission to this journal, you can choose to post your manuscript on SSRN, and it will be made publicly available as soon as it passes the journal's initial desk review.

As a preprint on SSRN, your manuscript will benefit from:

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Your decision to post or not post your preprint will have no effect on the editorial process or publication outcome with the journal. For additional information, please consult the [SSRN Terms of Use](#) and [FAQs](#).

It is expected that the corresponding author will seek approval from all co-authors before agreeing to post the manuscript publicly, prior to peer review, on SSRN.

Use of inclusive language

Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing which might imply one individual is superior to another on the grounds of:

- age
- gender
- race
- ethnicity
- culture
- sexual orientation
- disability or health condition

We recommend avoiding the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

No assumptions should be made about the beliefs of readers and writing should be free from bias, stereotypes, slang, reference to dominant culture and/or cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals or eukaryotic cells. This should be done in accordance with any requirements set by funders or sponsors and best practices within a field.
- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.
- Definitions of sex and/or gender applied should be explicitly stated to enhance the precision, rigor and reproducibility of the research and to avoid ambiguity or conflation of terms and the constructs to which they refer.

We advise you to read the [Sex and Gender Equity in Research \(SAGER\) guidelines](#) and the [SAGER checklist](#) (PDF) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting and research interpretation.

For further information we suggest reading the rationale behind and recommended [use of the SAGER guidelines](#).

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).
- Gender generally refers to socially constructed roles, behaviors and identities of women, men and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact and how power is distributed in society.

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

Image manipulation

We accept that authors sometimes need to adjust images for clarity but any manipulation of images for the purpose of deception or fraud will be seen as scientific ethical abuse and will be dealt with accordingly.

Authors must adhere to this journal's policy for graphical images:

- No specific feature within an image may be enhanced, obscured, moved, removed or introduced.
- Adjustments of brightness, contrast, or color balance are acceptable if, and only as long as, they do not obscure or eliminate any information present in the original image.
- Nonlinear adjustments such as changes to gamma settings must be disclosed in the figure legend.
- Use of AI tools to create or manipulate primary research images is not permitted. This includes images that represent experimental or observational data, such as western blots, microscopy images, histology images, or patient images. Please read our policy on the use of AI

tools in figures, images and artwork, which can be found in Elsevier's [GenAI Policies for Journals](#).

To verify compliance with the above, this journal may send your images to a secure third-party tool to enable screening for image irregularities. Our editors may ask you to provide original data or images if any questions arise as a result of the screening. The final decision as to whether images are acceptable will be taken by our editors.

Authors are encouraged to carefully check all images before submission and to connect all the data in any figures to the original, unprocessed data.

Jurisdictional claims

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- **Maps:** Readers should be able to locate any study areas shown within maps using common mapping platforms. Maps should only show the area actually studied and authors should not include a location map which displays a larger area than the bounding box of the study area. Authors should add a note clearly stating that "*map lines delineate study areas and do not necessarily depict accepted national boundaries*". During the review process, Elsevier's editors may request authors to change maps if these guidelines are not followed.
- **Institutional affiliations:** Authors should use either the full, standard title of their institution or the standard abbreviation of the institutional name so that the institutional name can be independently verified for research integrity purposes.

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If you need to delay the publication of your article for any reason, please contact the editorial office of the journal or our [Journal Article Publishing Support Center](#) at the earliest possible opportunity. While we will try to ensure that your article is not published before a certain date if you make your request on a timely basis, we cannot guarantee this. Please also note that once the agreed period of delay has expired the article may be published at any time thereafter.

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For questions about article withdrawal, correction, or retraction, please consult Elsevier's [Article Correction, Retraction and Removal Policy](#).

Writing and formatting

File format

We ask you to provide editable source files for your entire submission (including figures, tables and text graphics). Some guidelines:

- Save files in an editable format, using the extension .doc/.docx for Word files and .tex for LaTeX files. A PDF is not an acceptable source file.
- Format Word files in a single-column layout. Double-column formatting is only permitted for LaTeX submissions.
- Remove any strikethrough and underlined text from your manuscript, unless it has scientific significance related to your article.
- Use spell-check and grammar-check functions to avoid errors.

We advise you to read our [Step-by-step guide to publishing with Elsevier](#).

LaTeX

We encourage you use our [LaTeX template](#) when preparing a LaTeX submission. You will be asked to provide all relevant editable source files upon submission or revision.

Support for your LaTeX submission:

- [LaTeX submission instructions and templates](#)

- Journal Article Publishing Support Center [LaTeX FAQs and support](#)
- Researcher Academy's [Beginners' guide to writing a manuscript in LaTeX](#)

Title page

You are required to include the following details on the title page:

- Article title. Article titles should be concise and informative. Please avoid abbreviations and formulae, where possible, unless they are established and widely understood, e.g. DNA.
- Author names. Provide the given name(s) and family name(s) of each author. The order of authors should match the order in the submission system. Carefully check that all names are accurately spelled. You may add your name between parentheses in your own script after the English transliteration.
- Affiliations, referring to where the work was carried out, below the author names. Indicate affiliations using a lower-case superscript letter immediately after the author's name and in front of the corresponding address. Ensure that you provide the full name and postal address of each affiliation, including the country name.
- Corresponding author. Clearly indicate who will handle correspondence for your article at all stages of the refereeing and publication process and also post-publication. This responsibility includes answering any future queries about your results, data, or methodology. The email address of the corresponding author must be included on the title page and will be published in the article.
- Present/permanent address. If an author has changed affiliation since the work described in your article was carried out, or was visiting during that time, a "present address" (or "permanent address") can be indicated by a footnote to the author's name. The address where the author carried out the work must be retained as their main affiliation address. Use superscript Arabic numerals for such footnotes.

Abstract

You are required to provide a concise and factual abstract which does not exceed 250 words. The abstract should briefly state the purpose of your research, principal results and major conclusions. Some guidelines:

- Abstracts must be able to stand alone as abstracts are often presented separately from the article.
- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).
- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Please try to avoid keywords consisting of multiple words (using "and" or "of").

We recommend that you only use abbreviations in keywords if they are firmly established in the field.

Highlights

You are required to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
- Highlights should consist of 3 to 5 bullet points, each a maximum of 85 characters, including spaces.

We encourage you to view example [article highlights](#) and read about the benefits of their inclusion.

Graphical abstract

You are encouraged to provide a graphical abstract at submission.

The graphical abstract summarizes the contents of your article in a concise, pictorial, professional form, highlighting important aspects of the research. It is designed to appeal to an interdisciplinary audience, and draw more attention to your online article. Some guidelines:

- Ensure you have obtained the necessary permission to include any third party owned material in a graphical abstract.
- Use of generative AI or AI-assisted tools in the production of graphical abstracts must align with Elsevier's [GenAI Policies for Journals](#).
- Submit your graphical abstract as a separate file in the online submission system.
- Ensure the image is 531 x 1328 pixels (h x w) or proportionally more, and is readable at a size of 5 x 13 cm.
- Preferred file types for graphical abstracts are TIFF, EPS, PDF or MS Office files.

More information, examples and a template can be found in the [graphical abstract guidelines](#).

Math formulae

- Submit math equations as editable text, not as images.
- Present simple formulae in line with normal text, where possible.
- Use the solidus (/) instead of a horizontal line for small fractional terms such as X/Y.
- Present variables in italics.
- Denote powers of e by exp.
- Display equations separately from your text, numbering them consecutively in the order they are referred to within your text.

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

- Place tables next to the relevant text or on a separate page(s) at the end of your article.
- Cite all tables in the manuscript text.
- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.
- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

We recommend that you use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images and artwork

Figures, images, diagrams and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these [artwork and media instructions](#). Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.
- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text to a minimum, though any symbol or abbreviation should be explained.

Read how to add captions to your submission [here](#).

Formats

Submit your artwork in the formats listed below, following the given requirements:

Image type	Recommended file type	Requirements
Vector drawings	EPS, PDF	Embed the font or save text as graphics
Photographs (color or grayscale)	TIFF, JPG, PNG	Minimum resolution 300 dpi (min. width: single column: 1063 pixels; full page: 2244 pixels)
Bitmapped line drawings	TIFF, JPG, PNG	Minimum resolution 1000 dpi (min. width: single column: 3543 pixels; full page: 7480 pixels)
Charts and large or shaded tables	XLS, XLSX	Supply in a clearly readable format
Annotated images	PPT, PPTX	Annotations and text must be legible; Minimum resolution of embedded images 300 dpi

Do not submit:

- Files that are too low in resolution.
- Artwork where text is disproportionately small compared to the image size, as text may become unreadable.
- Multiple images or graphs combined into one file, as this affects legibility and accessibility. An exception may be made for a small number of images that share a direct relationship, such as before, during, and after photographs.

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